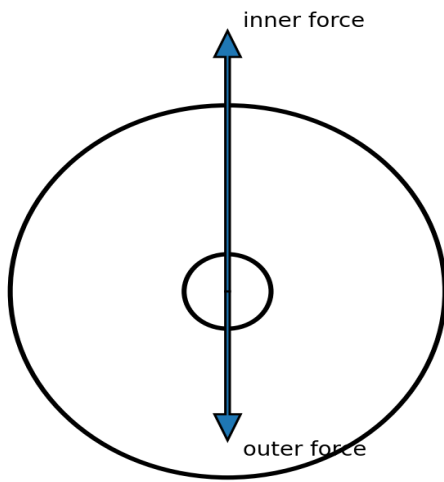


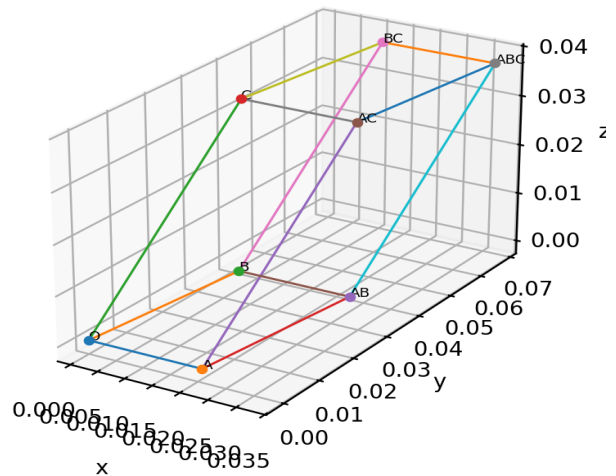
K-Field Concentric Counter-Rotating Rotor Lateral-Force Analysis

Equal-energy comparison, reaction-constrained spin-up, residual-force scaling, and complete computational reconstruction

Concentric counter-rotation



Base-cell geometry



P. Dwayne Esterline, Principal Investigator

BS 1999 Huntington University | MBA 2008 Indiana Wesleyan University
Krytronx

Generation UTC: 2026-07-27T17:04:18Z

Generation epoch: 1785171858

Machine-readable companion: K-Field_concentric_rotor_lateral_force_analysis_1785171858.json

PROPRIETARY CONFIDENTIAL INFORMATION AND TRADE SECRET OF KRYTRONX, LLC.

All Rights Reserved. Do Not Distribute.

Proprietary Notice and Document Contract

This report is an archival technical reference for the K-Field concentric-rotor force problem. It contains the complete governing assumptions, coordinate definitions, derivations, numerical examples, geometric reconstruction data, scaling laws, figures, and machine-readable reconstruction record needed to reproduce every reported value.

Classification. Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute. The analytical constants, geometric ledgers, force reductions, and device implications are maintained as controlled internal research information.

The report uses the calibrated axisymmetric K-Field rotor law stated in Section 4. Ordinary rigid-body mechanics supplies the independent constraints of rotational kinetic energy, angular momentum, torque reaction, and device center-of-mass acceleration. The central calculation compares those constraints without substituting one for another.

Central result: equal rotational kinetic energy does not require K-Field force cancellation. A reaction-driven concentric pair starting from zero angular momentum generally retains a lateral residual because internal torque balance cancels I times omega, whereas the adopted K-Field force law cancels only when m times omega cancels.

Field	Archival entry
Document title	K-Field Concentric Counter-Rotating Rotor Lateral-Force Analysis
Principal Investigator	P. Dwayne Esterline
Organization	Krytronx
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation UTC	2026-07-27T17:04:18Z
Generation epoch	1785171858
Output PDF	K-Field_concentric_rotor_lateral_force_analysis_1785171858.pdf
Machine-readable companion	K-Field_concentric_rotor_lateral_force_analysis_1785171858.json
PDF generation	Python ReportLab with built-in core fonts only
Figure generation	Python numerical computation and raster rendering from the exact report ledger

Contents

Proprietary Notice and Document Contract	2
1. Scope, Definitions, and Coordinate Lock	6
1.1 Objective	6
1.2 Adopted field vector and optimal orientation	6
1.3 Symbols and units	7
1.4 Separation of mechanical and K-Field balances	7
2. Base Cell Geometry and Phase Planes	7
2.1 Metric input	7
2.2 Canonical metric-preserving embedding	7
2.3 Complete node ledger	8
2.4 Gram matrix, plane areas, and normals	8
2.5 Role of the cell geometry in this rotor analysis	9
3. Mechanical Model of the Concentric Rotors	9
3.1 Axial inertia representation	9
3.2 Rotational energy and angular momentum	9
3.3 Reaction-only spin-up	9
4. Axisymmetric K-Field Rotor Law	10
4.1 Vector force law	10
4.2 Dimensional closure	10
4.3 Counter-rotating pair	11
5. Equal Rotational Energy with Unequal Mass and Speed	11
5.1 Equal-energy constraint	11
5.2 Equal-energy K-Field force ratio	11
5.3 Exact equal-energy cancellation boundary	11
5.4 Angular momentum of an equal-energy pair	12
6. Worked Equal-Energy Example	12
6.1 Input geometry and inner state	12

6.2 Equal-energy speed solution	13
6.3 K-Field forces	13
6.4 Why this state is not the reaction-only state	13
7. Reaction-Constrained Internal Spin-Up	14
7.1 Angular-momentum constraint	14
7.2 Residual K-Field force	14
7.3 Force ratio and cancellation condition	14
7.4 Energy partition under reaction-only operation	14
7.5 Compatibility theorem	14
8. Worked Reaction-Only Example	14
8.1 Outer speed required by internal reaction	14
8.2 Force and energy ledger	15
8.3 Direct comparison of the two outer speeds	15
9. General Scaling Laws and Design Maps	16
9.1 Reaction-constrained radius ratio	16
9.2 Absolute reaction force at fixed angular momentum	16
9.3 Mass, speed, and energy trade	17
9.4 Different rotor shape factors	18
10. Time-Domain Spin-Up, Dwell, and Spin-Down	18
10.1 Force during internal torque transfer	18
10.2 Exact one-second ramp to the worked state	19
10.3 Lateral impulse	19
10.4 Steady force versus transient torque	19
11. Whole-Assembly Force and Acceleration	19
11.1 External lateral force on the complete device	19
11.2 Directional control	20
11.3 Sign of the residual	20
12. Measurement Architecture and Falsifiable Scaling Tests	20
12.1 Required measured quantities	20

12.2 Primary discriminating tests 20

12.3 Control states 20

13. Findings and Design Consequences 21

13.1 Equal-energy finding 21

13.2 Reaction-only finding 21

13.3 Fundamental relation 21

13.4 Answer to the central question 21

Appendix A. Complete Numerical Input Ledger 21

Appendix B. Exact Recomputed Results 22

Appendix C. Selected Reaction Radius-Ratio Ledger 22

 C.1 Selected equal-energy mass-radius states 23

Appendix D. Reconstruction Procedure 23

Appendix E. Archival Record 23

1. Scope, Definitions, and Coordinate Lock

1.1 Objective

Consider two concentric rotating masses mounted on a common axis. The inner rotor is smaller and faster; the outer rotor is larger and slower. Their spin directions are opposite. The common spin axis is oriented for maximal K-Field lateral force. The objective is to determine whether the inner rotor can produce a larger lateral force than the opposing outer rotor when their rotational energies are equal, and then to determine the physically distinct state produced when the inner rotor is spun up by reacting internally against the outer rotor.

The two questions require two separate constraints. Equal rotational kinetic energy prescribes one relationship between speed, mass, radius, and inertia shape factor. Reaction-only spin-up from rest prescribes equal and opposite angular momentum. These constraints coincide only in a special equal-inertia state.

1.2 Adopted field vector and optimal orientation

The calibrated phase-velocity vector used throughout this report is

$$\begin{aligned} \mathbf{g} &= G_{\text{phase}} \mathbf{V}_E \\ &= (-880221.981635, -143911.436562, 266335.026949) \text{ m/s} \\ |\mathbf{g}| &= 930825.2172705192 \text{ m/s} \\ \mathbf{e}_g &= (-0.945636157362, -0.154606293310, 0.286127859460) \end{aligned}$$

Let $\mathbf{n}$ be the unit vector along the common spin axis. Maximal lateral-force magnitude occurs when $\mathbf{n} \cdot \mathbf{g}$ equals zero. For positive spin about $\mathbf{n}$, the force direction is parallel to $\mathbf{n} \times \mathbf{g}$. Reversing spin reverses the force by 180 degrees.

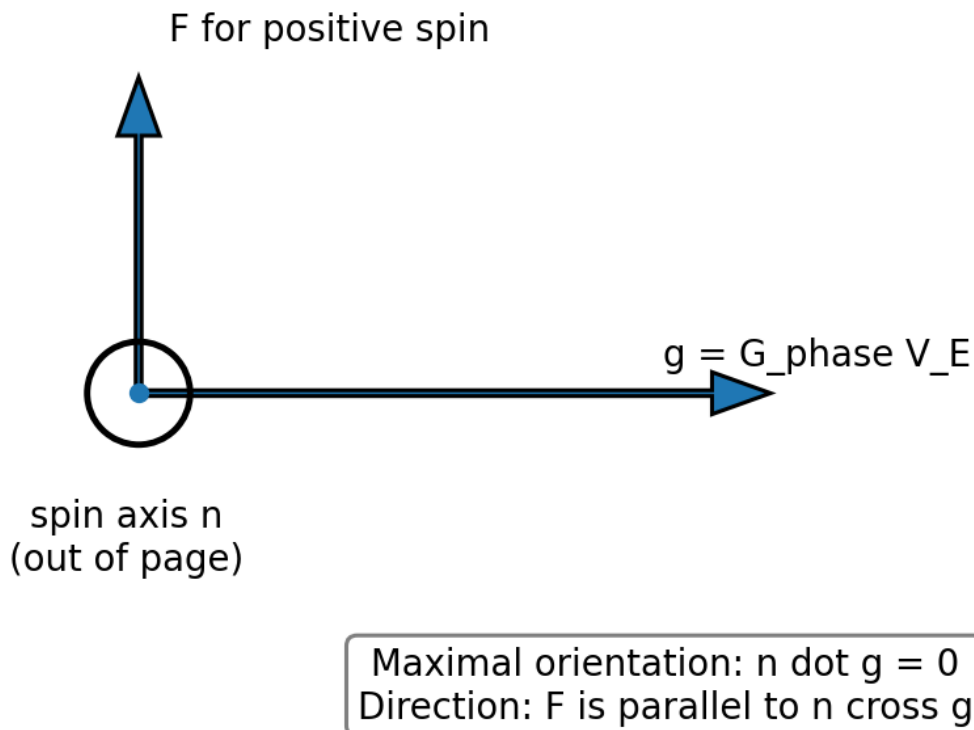


Figure 1. Maximal-orientation vector triad. The spin axis is perpendicular to $\mathbf{g}$, and the lateral-force direction follows the right-hand cross product $\mathbf{n} \times \mathbf{g}$. The outer rotor counter-rotates, so its force points oppositely.

1.3 Symbols and units

Symbol	Definition	SI unit
m_i, m_o	Inner and outer active rotating masses	kg
r_i, r_o	Effective rotational radii	m
beta_i, beta_o	Inertia shape factors in $I = \text{beta } m \text{ } r^2$	dimensionless
omega_i, omega_o	Signed angular speeds about the common axis	rad/s; rad is dimensionless
I_i, I_o	Axial moments of inertia	kg m^2
E_i, E_o	Rotational kinetic energies	J
L	Magnitude of equal and opposite rotor angular momentum in reaction-only operation	kg m^2/s
C_K	Maximal K-Field rotor coefficient, $\kappa_F g / 2$	N s/kg = m/s
F_i, F_o	Opposing K-Field force magnitudes	N
F_net	Signed net lateral force, positive in the inner-rotor force direction	N

1.4 Separation of mechanical and K-Field balances

Ordinary internal motor torque creates equal and opposite angular impulses on the two rotors. That balance controls $I \omega$. The K-Field law used here is proportional to $m \omega$. Because I equals $\text{beta } m \text{ } r^2$, equal angular momentum does not generally imply equal $m \omega$. The radius and inertia-shape factors remain in the residual force.

2. Base Cell Geometry and Phase Planes

2.1 Metric input

The base cell is defined by three direct basis-vector lengths and three included angles. The values below are retained in native source length units. No SI conversion is imposed because no native-to-SI length scale is required for the concentric-rotor force balance.

Metric	Value
a	0.0207970587 native length units
b	0.0364942052 native length units
c	0.0493085660 native length units
angle(a,b)	83.933483 deg
angle(a,c)	80.448130 deg
angle(b,c)	50.363232 deg
Area(a,b)	7.547217880965119e-04 native area units
Area(a,c)	1.011255755679060e-03 native area units
Area(b,c)	1.385784440909538e-03 native area units
Cell volume	2.842066945211215e-05 native volume units

2.2 Canonical metric-preserving embedding

A convenient local Cartesian embedding places a along positive x, places b in the x-y plane, and assigns c a positive z component. This construction preserves all three input lengths and all three included angles exactly. It fixes a local plotting frame; the rotor force orientation remains locked separately by the ICRS vector g.

```
a = (0.020797058700, 0.000000000000, 0.000000000000)
b = (0.003856816120, 0.036289833047, 0.000000000000)
c = (0.008182284384, 0.030762382380, 0.037657147177)
coordinates are in native length units
```

Canonical metric-preserving embedding of the base cell

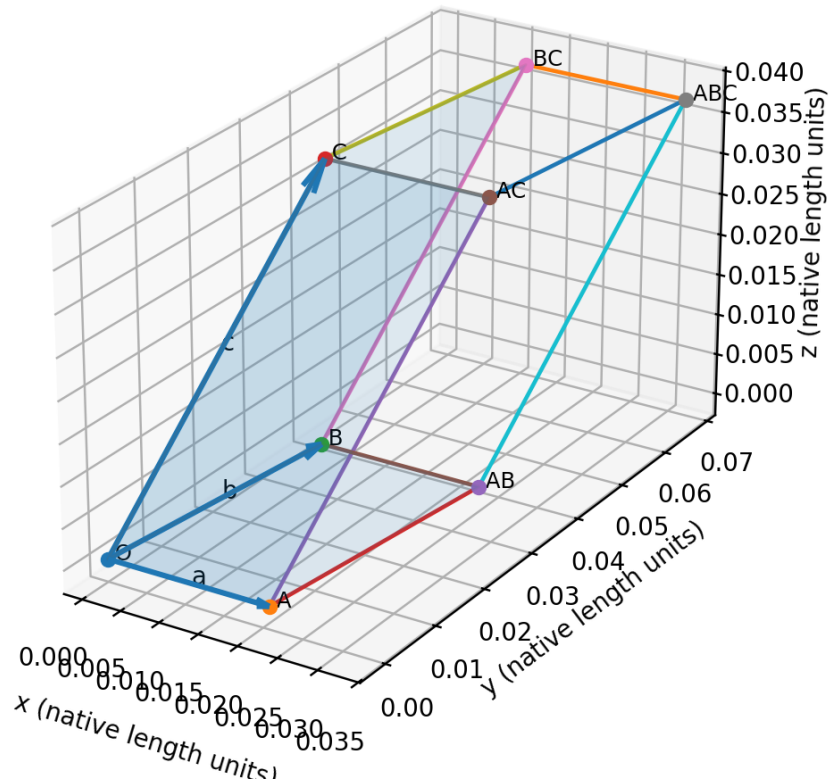


Figure 2. Dimensionally scaled canonical embedding of the complete base cell. All eight nodes are labeled. Axis scales are native length units, and the plotted edge coordinates are generated directly from the metric ledger.

2.3 Complete node ledger

Node	x	y	z	Vector definition
O	0.000000000000	0.000000000000	0.000000000000	0
A	0.020797058700	0.000000000000	0.000000000000	a
B	0.003856816120	0.036289833047	0.000000000000	b
C	0.008182284384	0.030762382380	0.037657147177	c
AB	0.024653874820	0.036289833047	0.000000000000	a+b
AC	0.028979343084	0.030762382380	0.037657147177	a+c
BC	0.012039100504	0.067052215428	0.037657147177	b+c
ABC	0.032836159204	0.067052215428	0.037657147177	a+b+c

2.4 Gram matrix, plane areas, and normals

The direct-space Gram matrix contains all pairwise metric products. Its diagonal entries are squared edge lengths; its off-diagonal entries are edge-length products multiplied by the corresponding angle cosine.

G_direct =

```
[4.325176505712e-04  8.021043124314e-05  1.701674486330e-04]
[8.021043124314e-05  1.331827013180e-03  1.147919287031e-03]
[1.701674486330e-04  1.147919287031e-03  2.431334680976e-03]
```

Plane	Area	Unit normal in canonical embedding	Four face nodes shown
a-b	7.547217880965119e-04	(0.000000000000, 0.000000000000, 1.000000000000)	O, A, B, AB
a-c	1.011255755679060e-03	(0.000000000000, -0.774440981837, 0.632646161492)	O, A, C, AC

Plane	Area	Unit normal in canonical embedding	Four face nodes shown
b-c	1.385784440909538e-03	(0.986135753693, -0.104804678115, -0.128655566277)	O, B, C, BC

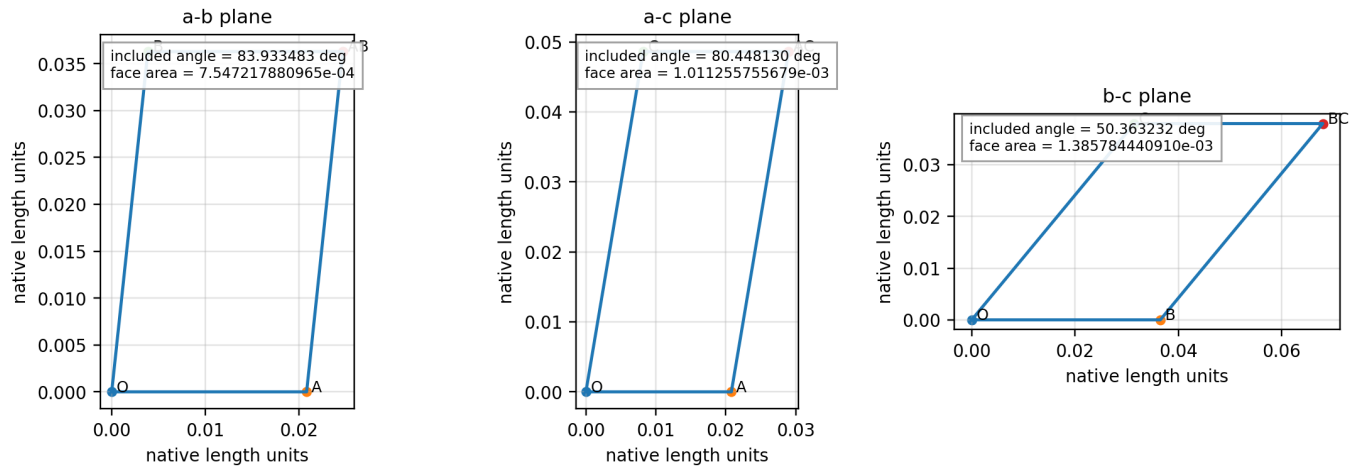


Figure 3. Dimensionally correct equal-aspect views of the three base planes. Each plot identifies all four nodes of the corresponding parallelogram, the included angle, and the computed face area.

2.5 Role of the cell geometry in this rotor analysis

The base cell and its phase planes define the geometric substrate from which the phase-response operator is constructed. For the present rotor reduction, that operator has already been evaluated against the Earth kinematic vector to produce $\mathbf{g}$. The mechanical balance therefore uses the calibrated vector $\mathbf{g}$ and coefficient C_K directly; no unreported cell scale or unreported projection factor is introduced.

3. Mechanical Model of the Concentric Rotors

3.1 Axial inertia representation

Each rotor is represented by an active rotating mass m , an effective radius r , and a shape factor β . The axial moment of inertia is

$$I = \beta m r^2 \quad (1)$$

Idealized rotor	β	Interpretation of r
Thin ring	1	Ring radius
Solid disk or solid cylinder about symmetry axis	1/2	Outer radius
Thin spherical shell	2/3	Sphere radius
Solid sphere	2/5	Sphere radius

The effective radius formulation keeps the derivation valid for different rotor geometries. A distributed rotor with known I can always be represented by βr^2 equal to I divided by m .

3.2 Rotational energy and angular momentum

$$E = (1/2) I \omega^2 = (1/2) \beta m r^2 \omega^2 \quad (2)$$

$$L = I \omega = \beta m r^2 \omega \quad (3)$$

Energy depends on the square of speed. Angular momentum depends linearly on speed and includes the full moment of inertia. Counter-rotation changes the sign of angular momentum but does not change the positive rotational kinetic energy.

3.3 Reaction-only spin-up

For an isolated two-rotor assembly initially at rest, the motor torque is internal. Equal and opposite angular impulses produce

$$\begin{aligned} I_i \omega_i + I_o \omega_o &= 0 \\ I_i \omega_i &= I_o |\omega_o| = L \end{aligned} \quad (4)$$

Equation (4) is the mechanical reaction constraint. It does not state that the rotor energies are equal. The smaller-inertia rotor necessarily acquires the larger angular speed.

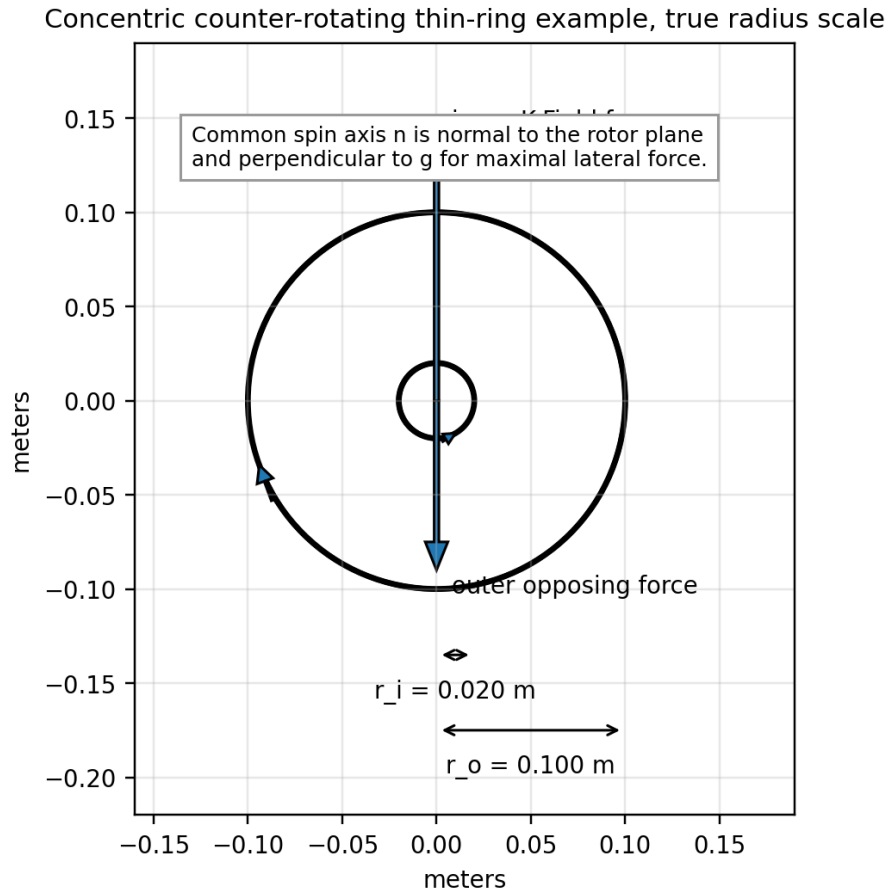


Figure 4. True-radius-scale cross-section of the worked concentric thin-ring pair. The inner radius is 0.020 m and the outer radius is 0.100 m. Counter-rotation reverses the K-Field force direction. Force-arrow lengths are schematic; exact values are reported in Sections 6 and 8.

4. Axisymmetric K-Field Rotor Law

4.1 Vector force law

For an axisymmetric active rotor, the adopted K-Field force law is

$$\mathbf{F}_{\mathbf{K_vector}} = (\kappa_F M / 2) (\boldsymbol{\Omega} \text{ cross } \mathbf{g}) \quad (5)$$

Here $\boldsymbol{\Omega}$ is the signed angular-velocity vector. At maximal orientation, $\boldsymbol{\Omega}$ is perpendicular to $\mathbf{g}$. The magnitude reduces to

$$\begin{aligned} F_{\mathbf{K}} &= C_{\mathbf{K}} M |\boldsymbol{\omega}| \\ C_{\mathbf{K}} &= \kappa_F |\mathbf{g}| / 2 = 7.5054\text{e-}6 \text{ N s/kg} \end{aligned} \quad (6)$$

Quantity	Value	Unit
$ \mathbf{g} $	930825.217270519235	m/s
$C_{\mathbf{K}}$	7.5054000000e-06	N s/kg = m/s
κ_F derived from $C_{\mathbf{K}}$ and $ \mathbf{g} $	1.612633577334e-11	dimensionless

4.2 Dimensional closure

In Equation (5), M has units kg, $\boldsymbol{\Omega}$ has units 1/s, and $\mathbf{g}$ has units m/s. Their product has units kg m/s squared, which is a newton. Consequently κ_F is dimensionless and $C_{\mathbf{K}}$ has units m/s, equivalently N s/kg.

4.3 Counter-rotating pair

Choose the positive lateral direction to be the force produced by the inner rotor. The outer rotor spins oppositely, so its K-Field force is negative in that coordinate. The net force is

$$F_{\text{net}} = C_K (m_i \omega_i - m_o |\omega_o|) \quad (7)$$

The pair cancels only when

$$m_i \omega_i = m_o |\omega_o| \quad (8)$$

Equation (8), not equal kinetic energy and not equal angular momentum, is the K-Field cancellation condition for the adopted axisymmetric rotor law.

5. Equal Rotational Energy with Unequal Mass and Speed

5.1 Equal-energy constraint

Set the inner and outer rotational kinetic energies equal while allowing different masses, radii, shape factors, and speeds:

$$\begin{aligned} (1/2) \beta_i m_i r_i^2 \omega_i^2 \\ = (1/2) \beta_o m_o r_o^2 |\omega_o|^2 \end{aligned} \quad (9)$$

Solving for the speed ratio gives

$$\begin{aligned} \omega_i / |\omega_o| \\ = (r_o/r_i) \sqrt{(\beta_o m_o)/(\beta_i m_i)} \end{aligned} \quad (10)$$

5.2 Equal-energy K-Field force ratio

Using F equals $C_K m \omega$ for both rotors and substituting Equation (10),

$$\begin{aligned} F_i/F_o \\ = (r_o/r_i) \sqrt{(\beta_o m_i)/(\beta_i m_o)} \end{aligned} \quad (11)$$

The smaller inner rotor produces more lateral force than the larger outer rotor when

$$m_i/(\beta_i r_i^2) > m_o/(\beta_o r_o^2) \quad (12)$$

For identical rotor types, β_i equals β_o , and the condition reduces to

$$m_i/m_o > (r_i/r_o)^2 \quad (13)$$

Thus the inner rotor may have substantially less mass than the outer rotor and still dominate if its radius is sufficiently smaller.

5.3 Exact equal-energy cancellation boundary

At equal energy, K-Field force cancellation occurs when the inequality in Equation (12) becomes equality:

$$m_i/(\beta_i r_i^2) = m_o/(\beta_o r_o^2) \quad (14)$$

For identical rotor types, the cancellation boundary is m_i/m_o equals the square of r_i/r_o . Above that boundary the inner rotor dominates; below it the outer rotor dominates.

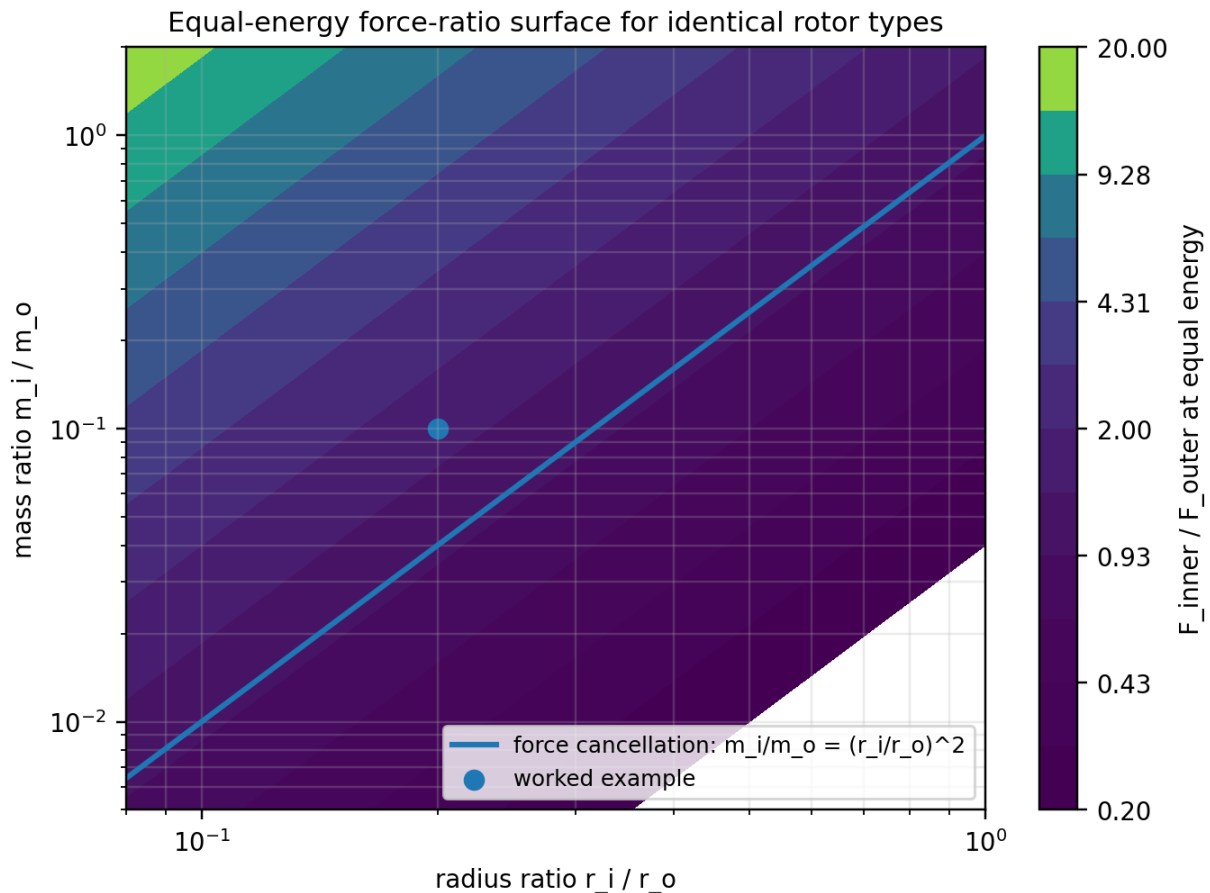


Figure 5. Equal-energy force-ratio surface for identical rotor types. The plotted boundary is the exact force-cancellation condition. The worked example lies above the boundary and therefore produces a residual in the inner-rotor force direction.

5.4 Angular momentum of an equal-energy pair

For a rotor carrying energy E , the angular-momentum magnitude is $\sqrt{2 E I}$. Therefore, two counter-rotating rotors with equal energy have zero total angular momentum only when their inertias are equal:

$$\begin{aligned} |L_i| &= \sqrt{2 E I_i} \\ |L_o| &= \sqrt{2 E I_o} \\ L_i + L_o &= 0 \text{ only if } I_i = I_o \end{aligned} \quad (15)$$

If the inertias differ, an equal-energy state with opposite spin directions contains nonzero total angular momentum. Reaching that state from a stationary isolated assembly requires angular momentum exchange with the housing, a support, or another external reservoir.

6. Worked Equal-Energy Example

6.1 Input geometry and inner state

The numerical example uses two thin rings, so β_i and β_o both equal one. The inner ring is ten times lighter and five times smaller in radius than the outer ring.

Parameter	Inner rotor	Outer rotor
Mass	0.100000 kg	1.000000 kg
Radius	0.020000 m	0.100000 m
Shape factor β	1.000000	1.000000
Moment of inertia	4.000000000000e-05 kg m ²	1.000000000000e-02 kg m ²
Specified speed	100000.000000 rpm	Solved by equal-energy condition

6.2 Equal-energy speed solution

```

omega_i = 10471.975511965977 rad/s
E_i = (1/2) I_i omega_i^2 = 2193.245422464302 J
|omega_o| = sqrt(2 E_i / I_o) = 662.305884386407 rad/s
|rpm_o| = 6324.555320336757 rpm
E_o = 2193.245422464302 J

```

6.3 K-Field forces

```

F_i = C_K m_i omega_i = 7.859636500751e-03 N = 7.859637 mN
F_o = C_K m_o |omega_o| = 4.970870584674e-03 N = 4.970871 mN
F_net = F_i - F_o = 2.888765916077e-03 N = 2.888766 mN

```

Quantity	Inner rotor	Outer rotor	Pair result
Rotational energy	2193.245422 J	2193.245422 J	Equal by construction
Speed	100000.000000 rpm	6324.555320 rpm	ratio 15.811388301
K-Field force magnitude	7.859637 mN	4.970871 mN	net 2.888766 mN
Angular momentum magnitude	0.418879020 kg m ² /s	6.623058844 kg m ² /s	signed total -6.204179823 kg m ² /s

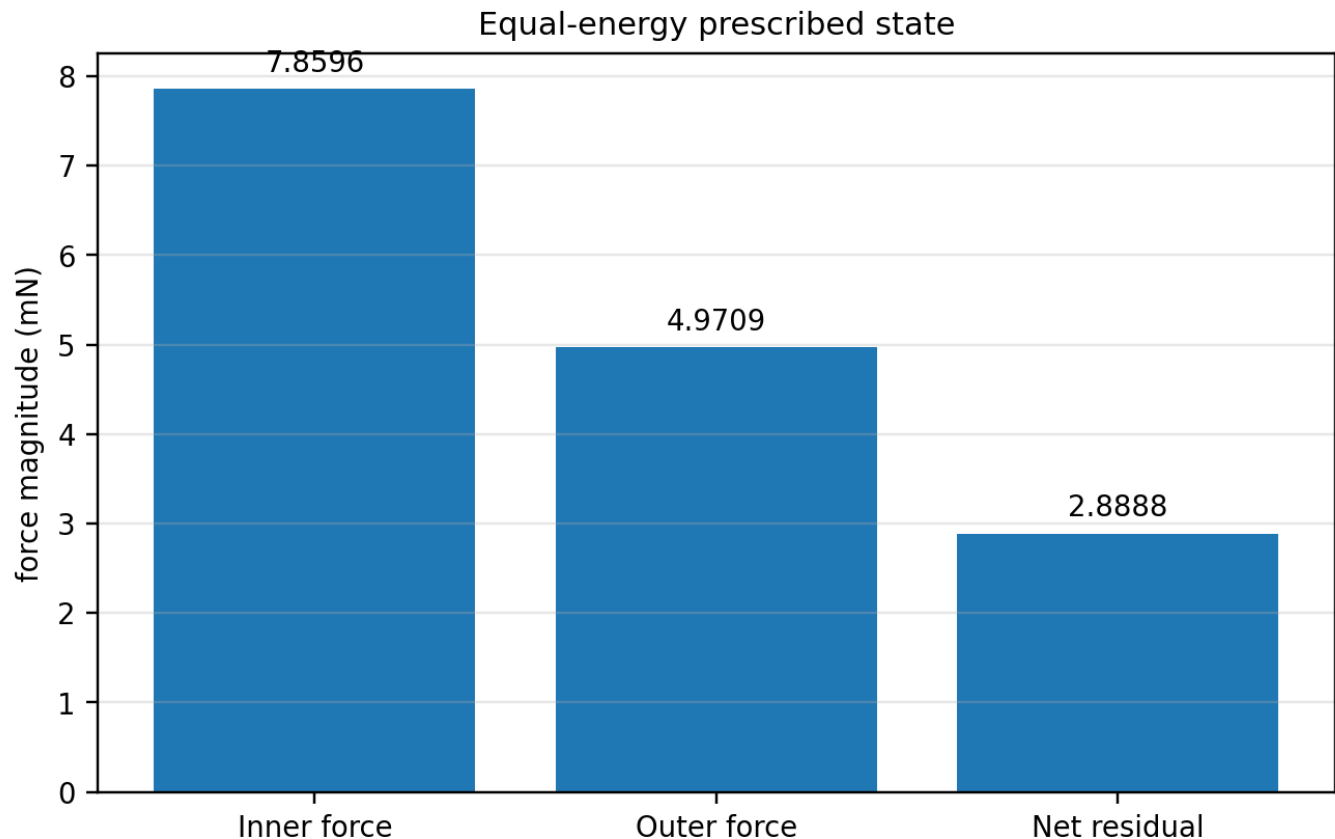


Figure 6. Computed K-Field forces for the equal-energy prescribed state. Equal rotational energy leaves a 2.888766 mN lateral residual in the inner-rotor force direction.

6.4 Why this state is not the reaction-only state

The equal-energy state carries inner angular momentum 0.418879020 kg m squared per second and opposing outer angular momentum 6.623058844 kg m squared per second. Their signed sum is -6.204179823 kg m squared per second. An isolated assembly initially at rest cannot reach this state through internal torque alone because its total angular momentum would no longer be zero.

Equal-energy result: $F_{\text{net}} = 2.888766 \text{ mN}$. Required net external angular impulse from a zero-angular-momentum start: $-6.204179823 \text{ kg m}^2/\text{s}$, with sign set by the selected spin convention.

7. Reaction-Constrained Internal Spin-Up

7.1 Angular-momentum constraint

Now impose the actual reaction-wheel condition. The inner rotor is accelerated by a motor that reacts against the outer rotor. Starting from rest with no external axial torque,

$$I_i \omega_i = I_o |\omega_o| = L \quad (16)$$

The resulting speed ratio is

$$\begin{aligned} \omega_i / |\omega_o| \\ = (\beta_o m_o r_o^2) / (\beta_i m_i r_i^2) \end{aligned} \quad (17)$$

7.2 Residual K-Field force

Substituting $m \omega$ equals L divided by βr squared into the K-Field force law gives

$$\begin{aligned} F_i &= C_K L / (\beta_i r_i^2) \\ F_o &= C_K L / (\beta_o r_o^2) \end{aligned} \quad (18)$$

Therefore the net force is

$$F_{\text{net}} = C_K L [1/(\beta_i r_i^2) - 1/(\beta_o r_o^2)] \quad (19)$$

At fixed transferred angular momentum L , the rotor masses cancel from Equation (19). Mass controls the speeds and energy required to carry L , but the residual force is controlled by the effective rotational radii βr squared.

7.3 Force ratio and cancellation condition

$$F_i / F_o = (\beta_o r_o^2) / (\beta_i r_i^2) \quad (20)$$

$$F_{\text{net}} = 0 \text{ only if } \beta_i r_i^2 = \beta_o r_o^2 \quad (21)$$

For identical rotor types, the force ratio is simply the square of the radius ratio. A smaller inner radius therefore produces the larger force in the reaction-constrained pair.

7.4 Energy partition under reaction-only operation

The two rotors carry equal angular-momentum magnitudes but generally unequal energies:

$$\begin{aligned} E_i &= L^2 / (2 I_i) \\ E_o &= L^2 / (2 I_o) \\ E_i / E_o &= I_o / I_i = \omega_i / |\omega_o| \end{aligned} \quad (22)$$

The smaller-inertia rotor stores more energy and spins faster. This is the ordinary mechanical cost associated with producing the K-Field asymmetry through unequal radii.

7.5 Compatibility theorem

Suppose equal energy and reaction balance are both imposed. Equal angular-momentum magnitude gives E equals L squared divided by $2I$. Equal energies therefore require I_i equals I_o . Equation (16) then requires equal speed magnitudes.

Consequently, equal energy, zero total angular momentum, and unequal speed cannot all hold simultaneously.

A high-rpm inner rotor and slow outer reaction rotor is necessarily an unequal-energy state when the assembly starts from rest and exchanges no angular momentum with its surroundings.

8. Worked Reaction-Only Example

8.1 Outer speed required by internal reaction

Keep the same inner ring at 100,000 rpm. Its angular momentum is transferred in equal and opposite amount to the outer ring.

$$\begin{aligned} L = I_i \omega_i &= 0.418879020479 \text{ kg m}^2/\text{s} \\ |\omega_o| &= L / I_o = 41.887902047864 \text{ rad/s} \end{aligned}$$

$|rpm_o| = 400.000000000000 \text{ rpm}$

8.2 Force and energy ledger

Quantity	Inner rotor	Outer rotor	Pair result
Speed	100000.000000 rpm	400.000000 rpm	ratio 250.000000
Angular momentum magnitude	0.418879020 kg m^2/s	0.418879020 kg m^2/s	signed total 0
Rotational energy	2193.245422464 J	8.772981690 J	ratio 250.000000
K-Field force	7.859637 mN	0.314385 mN	net 7.545251 mN

$F_i/F_o = (r_o/r_i)^2 = (0.100/0.020)^2 = 25.000000$
 $F_{net} = 7.545251040721e-03 \text{ N} = 7.545251 \text{ mN}$

Reaction-only result: the inner rotor produces 25 times the opposing outer-rotor force. The net steady lateral residual is 7.545251 mN in the inner-rotor force direction.

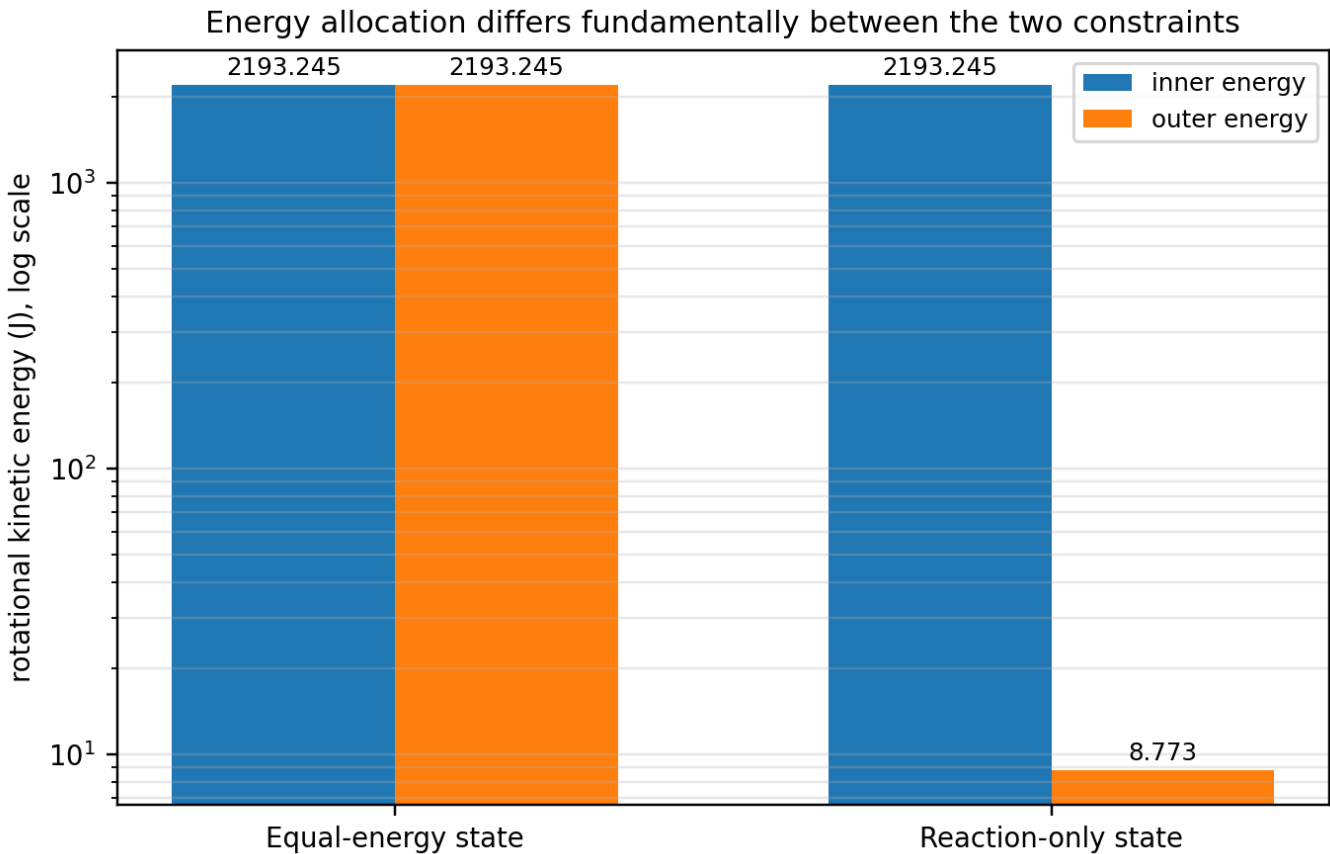


Figure 7. Energy allocation in the two distinct constraint states. The equal-energy state assigns the same 2193.245 J to both rotors but violates zero total angular momentum. The reaction-only state preserves zero total angular momentum and assigns only 8.772982 J to the outer rotor.

8.3 Direct comparison of the two outer speeds

State	Outer speed	Outer energy	Outer force	Net force
Equal energy	6324.555320 rpm	2193.245422 J	4.970871 mN	2.888766 mN
Reaction only	400.000000 rpm	8.772982 J	0.314385 mN	7.545251 mN

The reaction-only outer speed is much lower because the outer moment of inertia is 250 times the inner moment of inertia. Equal angular momentum therefore requires a 250-to-1 speed ratio. Equal energy requires only the square root of the inertia ratio, producing a 15.811-to-1 speed ratio.

9. General Scaling Laws and Design Maps

9.1 Reaction-constrained radius ratio

For identical rotor types define q equals r_o divided by r_i . Equations (20) and (19) become

$$\begin{aligned} F_i/F_o &= q^2 \\ F_{\text{net}}/F_i &= 1 - 1/q^2 \\ (F_i - F_o)/(F_i + F_o) &= (q^2 - 1)/(q^2 + 1) \end{aligned} \quad (23)$$

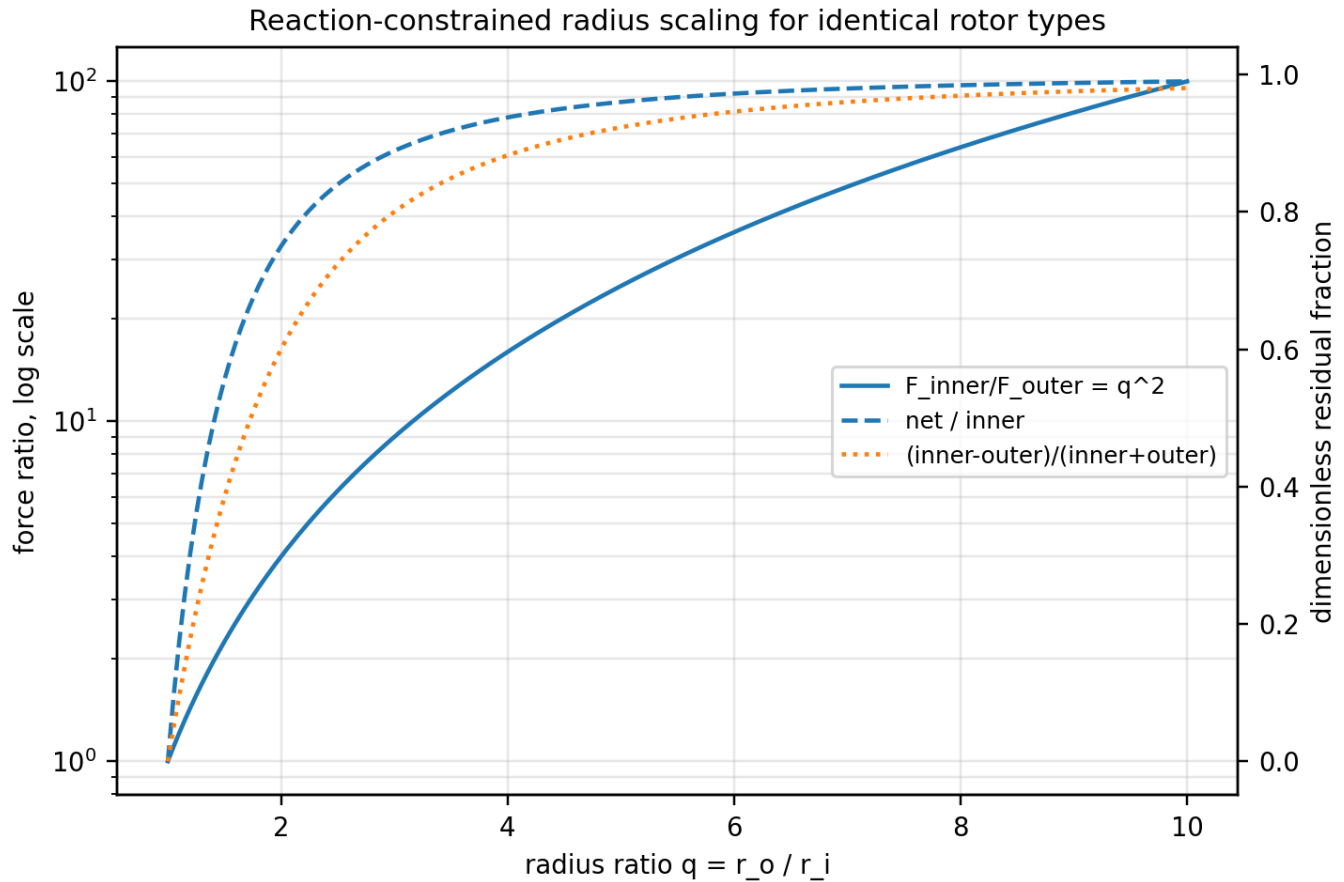


Figure 8. Reaction-constrained force ratio and normalized residual measures as functions of the outer-to-inner radius ratio. The force ratio grows quadratically, while the net force rapidly approaches the inner-rotor force as q increases.

9.2 Absolute reaction force at fixed angular momentum

For fixed geometry, Equation (19) is linear in transferred angular momentum. The sample pair has coefficient

$$\begin{aligned} F_{\text{net}}/L &= C_K [1/r_i^2 - 1/r_o^2] \\ &= 1.801296000000e-02 \text{ N per (kg m}^2/\text{s)} \end{aligned}$$

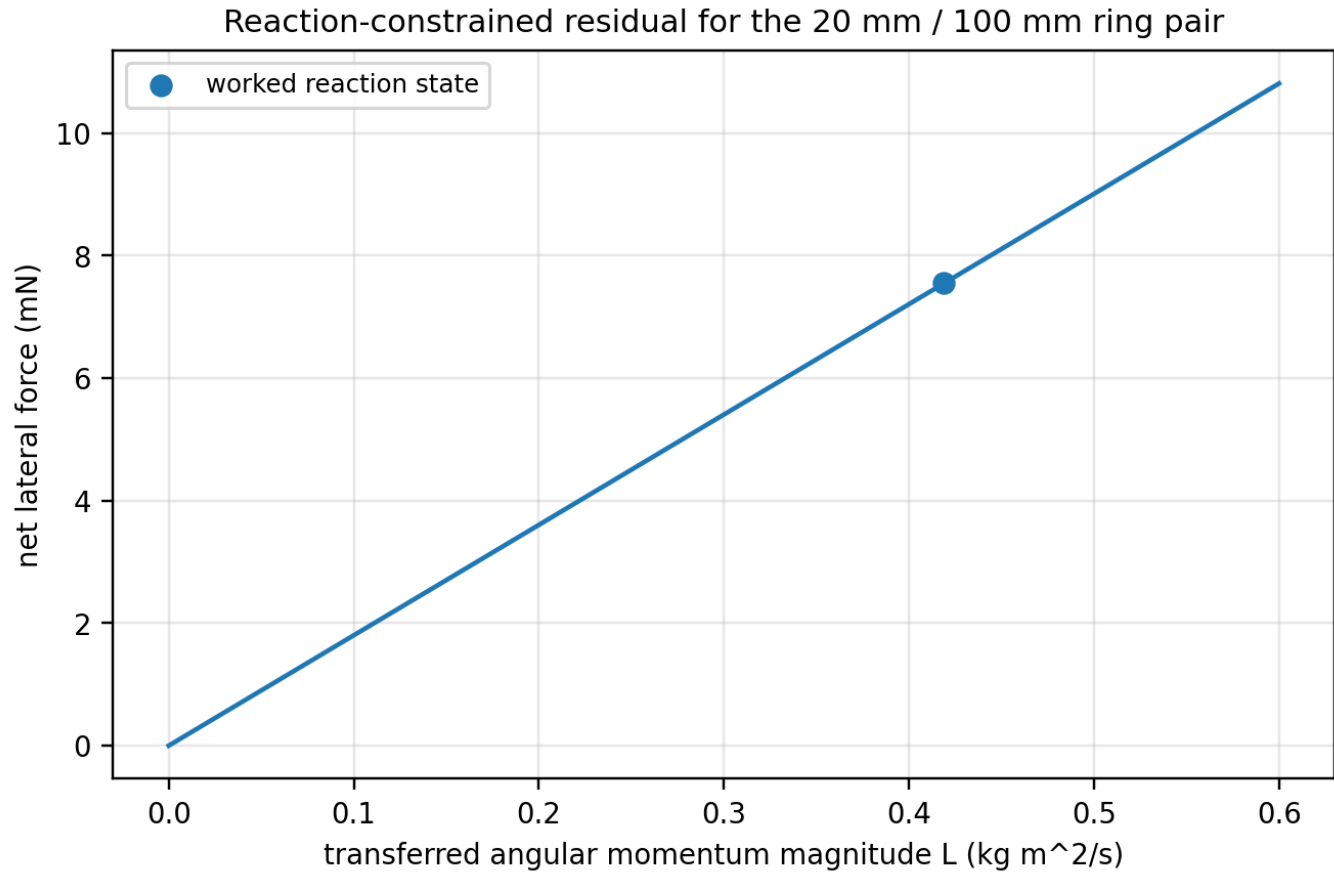


Figure 9. Net reaction-constrained lateral force versus transferred angular momentum for the worked 20 mm and 100 mm thin-ring geometry. The exact worked state is marked at $L = 0.418879 \text{ kg m squared per second}$.

9.3 Mass, speed, and energy trade

At fixed L and fixed effective radius, force is independent of rotor mass, but speed varies inversely with mass and energy varies inversely with inertia:

$$\begin{aligned}
 \omega &= L / (\beta m r^2) \\
 E &= L^2 / (2 \beta m r^2) \\
 F_K &= C_K L / (\beta r^2)
 \end{aligned}
 \tag{24}$$

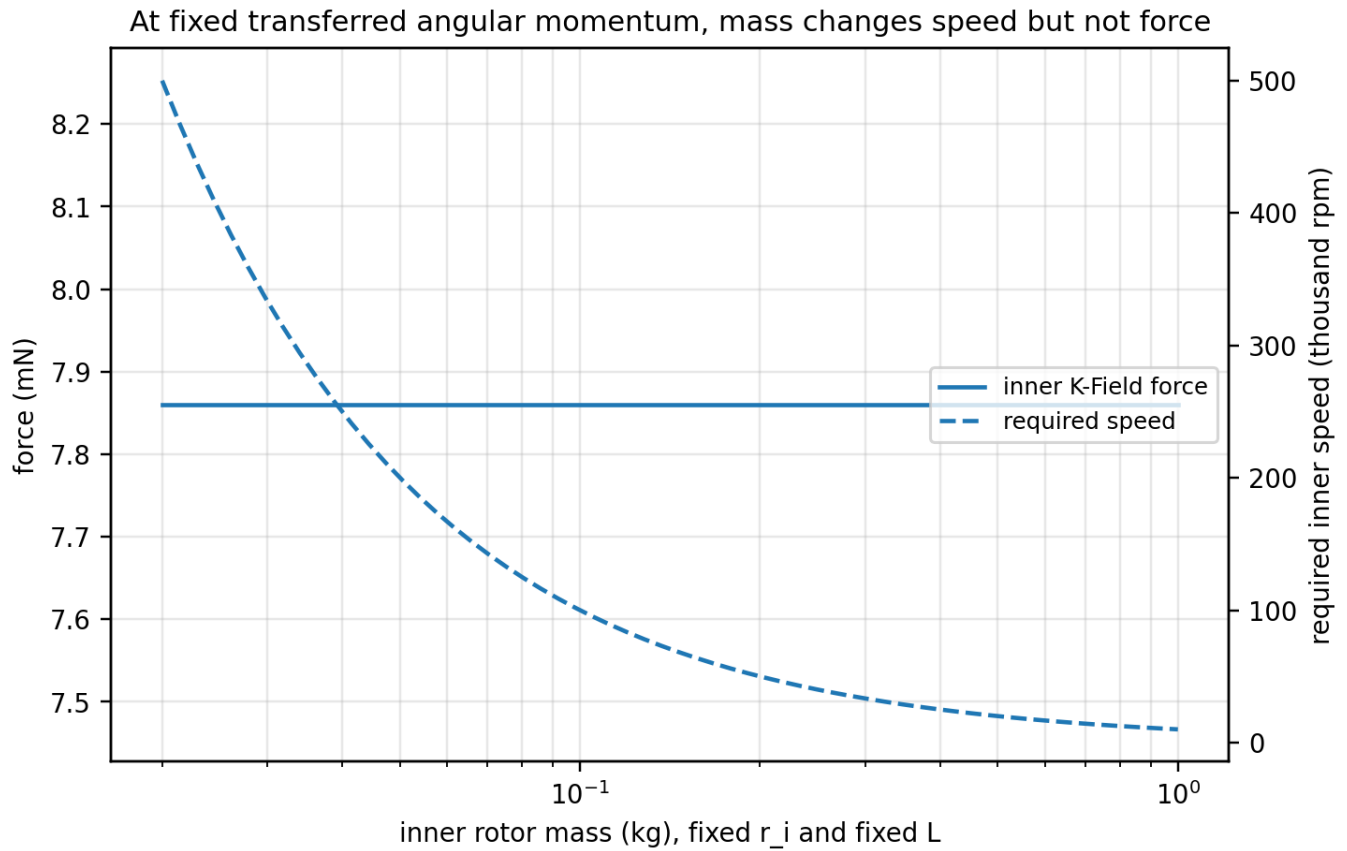


Figure 10. Inner-rotor force and required speed versus inner mass at fixed transferred angular momentum and fixed 0.020 m radius. The force remains constant; reducing mass requires proportionally higher speed.

9.4 Different rotor shape factors

If the inner and outer rotors have different mass distributions, the effective quantities $\beta_i r_i$ squared and $\beta_o r_o$ squared replace the simple squared radii. A solid disk at radius r has β equals one half, so at fixed L it produces twice the K-Field force of a thin ring at the same outer radius. The reason is that the disk must rotate twice as fast to carry the same angular momentum.

Condition	General result	Identical rotor types
Equal-energy force ratio	$(r_o/r_i) \sqrt{(\beta_o m_i)/(\beta_i m_o)}$	$(r_o/r_i) \sqrt{m_i/m_o}$
Equal-energy cancellation	$m_i/(\beta_i r_i^2) = m_o/(\beta_o r_o^2)$	$m_i/m_o = (r_i/r_o)^2$
Reaction force ratio	$\beta_o r_o^2/(\beta_i r_i^2)$	$(r_o/r_i)^2$
Reaction cancellation	$\beta_i r_i^2 = \beta_o r_o^2$	$r_i = r_o$

10. Time-Domain Spin-Up, Dwell, and Spin-Down

10.1 Force during internal torque transfer

Let $\tau(t)$ be the internal motor torque magnitude. The transferred angular momentum is the time integral of torque:

$$L(t) = \int_0^t \tau(\tau_{\text{prime}}) d\tau_{\text{prime}}$$

$$F_{\text{net}}(t) = C_K L(t) [1/(\beta_i r_i^2) - 1/(\beta_o r_o^2)] \quad (25)$$

For a constant torque, both L and the net K-Field force rise linearly during spin-up. Once the rotor speeds are held constant, L and the lateral residual remain constant in the ideal model.

10.2 Exact one-second ramp to the worked state

The worked reaction state requires $L = 0.418879020479$ kg m squared per second. A one-second linear spin-up therefore requires constant internal torque magnitude 0.418879020479 N m. The plotted sequence ramps from rest to the worked state in one second, dwells for 0.5 seconds, and returns to rest in one second.

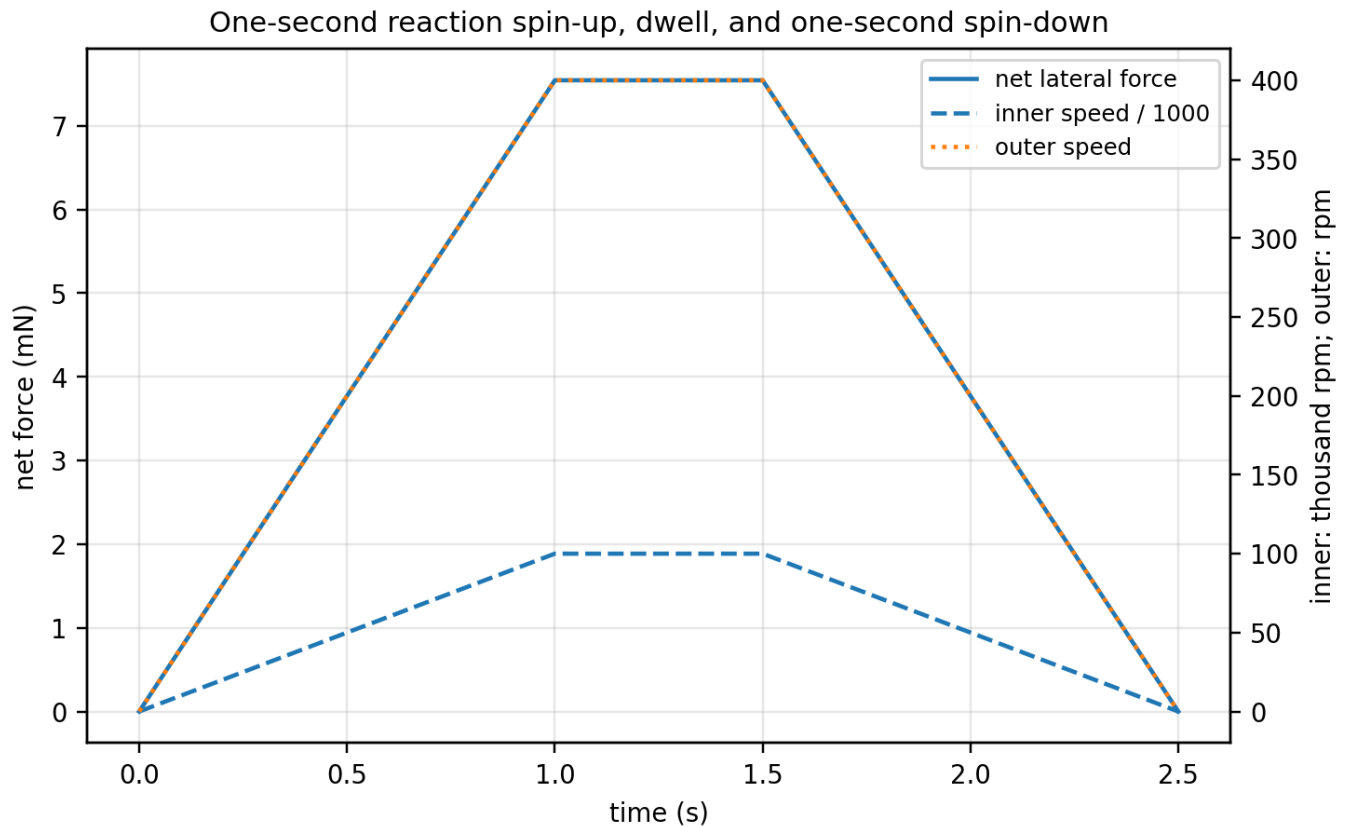


Figure 11. Computed force and speed history for a one-second reaction spin-up, 0.5-second dwell, and one-second spin-down. The force follows transferred angular momentum exactly.

10.3 Lateral impulse

The lateral impulse over any operating interval is the time integral of the net force:

$$J_{\text{lateral}} = \int F_{\text{net}}(t) dt = C_K [1/(\beta_i r_i^2) - 1/(\beta_o r_o^2)] \int L(t) dt \quad (26)$$

For the plotted symmetric ramp-dwell-ramp profile, the computed positive lateral impulse is $1.131787656108 \times 10^{-2}$ N s because the force retains the same lateral sign throughout spin-up, dwell, and spin-down. A subsequent cycle with both rotor spin directions reversed would produce the same impulse magnitude in the opposite lateral direction.

10.4 Steady force versus transient torque

The internal motor torque is required only to change the rotor speeds or compensate losses. The steady K-Field force in Equation (19) depends on the stored angular momentum state, not on continuing angular acceleration. Mechanical bearings and motor controls must nevertheless transmit the ordinary torque and radial loads required to maintain the concentric geometry.

11. Whole-Assembly Force and Acceleration

11.1 External lateral force on the complete device

The motor torque and bearing reactions are internal to the complete device. They cancel in the ordinary translational force balance. The K-Field forces in Equation (7) are externally directed field-coupled forces. Their vector sum is therefore the force available to accelerate the complete assembly.

$$a_{CM} = F_{net} / M_{device} \tag{27}$$

M_device includes both rotating masses and every passive component that translates with them: motor, bearings, shafts, housing, instrumentation, and structure. Passive mass reduces acceleration but does not alter the active-rotor force difference in Equation (19).

11.2 Directional control

For maximal magnitude, the spin axis remains perpendicular to g. Rotating the common axis around g rotates the lateral-force direction around the plane normal to g. Reversing both rotor spin directions reverses the residual force without changing either energy or angular-momentum magnitudes.

11.3 Sign of the residual

When $\beta_i r_i^2$ is smaller than $\beta_o r_o^2$, the inner rotor force is larger and the residual follows the inner rotor spin sign. If the effective radii are exchanged so that $\beta_i r_i^2$ is larger, the outer rotor dominates and the residual reverses. If the effective radii are equal, the reaction-constrained forces cancel exactly even when the rotor masses and speeds differ.

12. Measurement Architecture and Falsifiable Scaling Tests

12.1 Required measured quantities

A complete test record requires rotor masses, effective moments of inertia, radii, signed rotational speeds, common-axis orientation, motor torque or angular-momentum transfer history, full device mass, and the measured lateral support force. Direct tachometry on both rotors is necessary because equal-energy and reaction-only states can have very different outer speeds.

Measurement	Purpose	Model reconstruction
Inner and outer rpm with sign	Establish ω_i and ω_o	Direct evaluation of Equation (7)
Moments of inertia	Distinguish energy and angular-momentum constraints	$E = I \omega^2/2$; $L = I \omega$
Common-axis orientation relative to g	Verify maximal or swept orientation	Magnitude proportional to $ n \times g $
Lateral force or support reaction	Primary observable	Compare with F_{net}
Motor torque versus time	Reconstruct transferred angular momentum	$L(t) = \int \text{torque } dt$
Spin reversal	Verify vector sign reversal	F_{net} changes sign
Radius-ratio series	Verify quadratic reaction scaling	$F_i/F_o = \beta_o r_o^2/(\beta_i r_i^2)$

12.2 Primary discriminating tests

Constraint test. Prepare one prescribed equal-energy state and one reaction-only state using the same rotor geometry and inner speed. The outer speeds must be 6324.555 rpm and 400 rpm, respectively, for the worked pair. Their predicted residuals are 2.888766 mN and 7.545251 mN.

Radius test. Hold transferred angular momentum fixed while changing outer-to-inner effective radius ratio. For identical rotor types the force ratio must follow q^2 .

Mass test. Hold L and effective radii fixed while changing active rotor mass and compensating speed. Equation (24) predicts unchanged force and inversely varying rpm.

Reversal test. Reverse both spin directions at fixed axis orientation. The lateral force must reverse immediately in sign while retaining magnitude after the same speed magnitudes are restored.

12.3 Control states

The zero-speed state establishes the force baseline. Equal effective radii establish the reaction-cancellation control. Axis alignment parallel to g establishes the orientation null. These controls independently test speed dependence, reaction-radius dependence, and vector orientation.

13. Findings and Design Consequences

13.1 Equal-energy finding

Two counter-rotating rotors with equal rotational kinetic energy do not generally cancel. Their force ratio is set by Equation (11). In the worked case, the 0.10 kg inner ring at 20 mm radius and 100,000 rpm produces 7.859637 mN. The 1.00 kg outer ring at 100 mm radius requires 6324.555 rpm for equal energy and produces 4.970871 mN. The net residual is 2.888766 mN toward the inner-rotor force direction.

13.2 Reaction-only finding

When the inner rotor is spun up by reacting internally against the outer rotor, the mechanical constraint is equal and opposite angular momentum. The outer speed is then 400 rpm, not 6324.555 rpm. The opposing outer K-Field force is only 0.314385 mN, leaving a net residual of 7.545251 mN.

13.3 Fundamental relation

For reaction-only operation, the complete result is

$$F_{\text{net}} = C_K L \left[\frac{1}{(\beta_i r_i^2)} - \frac{1}{(\beta_o r_o^2)} \right]$$

This relation shows why the residual survives internal reaction. Ordinary torque balance removes the mass factors through L equals $\beta m r^2 \omega$. The K-Field source retains the inverse effective-radius-squared terms. Smaller effective radius therefore yields larger lateral force at the same transferred angular momentum.

13.4 Answer to the central question

Yes. Under the adopted K-Field rotor law, spinning the inner rotor by reacting off a larger concentric outer rotor produces a net lateral residual whenever $\beta_i r_i^2$ differs from $\beta_o r_o^2$. For the worked thin-ring pair, the internally reaction-driven residual is 7.545251 mN.

Appendix A. Complete Numerical Input Ledger

Input	Value	Unit or note
g_x	-880221.981634999975	m/s
g_y	-143911.436561999988	m/s
g_z	266335.026949000021	m/s
g	930825.217270519235	m/s
C_K	7.505400000000e-06	N s/kg
kappa_F	1.612633577334e-11	dimensionless
m_i	0.100000000000	kg
r_i	0.020000000000	m
beta_i	1.000000000000	dimensionless
rpm_i	100000.000000000000	rpm
m_o	1.000000000000	kg
r_o	0.100000000000	m
beta_o	1.000000000000	dimensionless
Base cell a	0.020797058700	native length units
Base cell b	0.036494205200	native length units
Base cell c	0.049308566000	native length units
Base cell angle(a,b)	83.933483000000	deg
Base cell angle(a,c)	80.448130000000	deg
Base cell angle(b,c)	50.363232000000	deg

Appendix B. Exact Recomputed Results

Result	Value	Unit
I_i	4.000000000000000e-05	kg m^2
I_o	1.000000000000000e-02	kg m^2
omega_i	1.047197551196598e+04	rad/s
E_i	2.193245422464302e+03	J
Equal-energy omega_o	6.623058843864067e+02	rad/s
Equal-energy rpm_o	6.324555320336757e+03	rpm
Equal-energy F_i	7.859636500750946e-03	N
Equal-energy F_o	4.970870584673737e-03	N
Equal-energy F_net	2.888765916077209e-03	N
Equal-energy L_i	4.188790204786391e-01	kg m^2/s
Equal-energy L_o	6.623058843864068e+00	kg m^2/s
Equal-energy signed L_total	-6.204179823385429e+00	kg m^2/s
Reaction omega_o	4.188790204786390e+01	rad/s
Reaction rpm_o	3.999999999999999e+02	rpm
Reaction E_o	8.772981689857207e+00	J
Reaction F_o	3.143854600300378e-04	N
Reaction F_net	7.545251040720908e-03	N
Reaction force ratio	2.500000000000001e+01	dimensionless
Reaction energy ratio	2.500000000000000e+02	dimensionless
One-second ramp torque	4.188790204786391e-01	N m
Ramp-dwell-ramp impulse	1.131787656108136e-02	N s

Appendix C. Selected Reaction Radius-Ratio Ledger

q = r_o/r_i	F_i/F_o	F_net/F_i	Asymmetry index
1.0000	1.00000000	0.00000000	0.00000000
1.2500	1.56250000	0.36000000	0.21951220
1.5000	2.25000000	0.55555556	0.38461538
2.0000	4.00000000	0.75000000	0.60000000
3.0000	9.00000000	0.88888889	0.80000000
4.0000	16.00000000	0.93750000	0.88235294
5.0000	25.00000000	0.96000000	0.92307692
7.5000	56.25000000	0.98222222	0.96506550
10.0000	100.00000000	0.99000000	0.98019802

C.1 Selected equal-energy mass-radius states

r_i/r_o	m_i/m_o	F_i/F_o	Dominant rotor
0.100000	0.005000	0.707106781	outer
0.100000	0.010000	1.000000000	cancellation
0.100000	0.100000	3.162277660	inner
0.200000	0.010000	0.500000000	outer
0.200000	0.040000	1.000000000	cancellation
0.200000	0.100000	1.581138830	inner
0.500000	0.100000	0.632455532	outer
0.500000	0.250000	1.000000000	cancellation
0.500000	1.000000	2.000000000	inner
1.000000	1.000000	1.000000000	cancellation

Appendix D. Reconstruction Procedure

The machine-readable companion contains the complete constants, vectors, cell nodes, equations, worked states, and sweep arrays. The following sequence reconstructs the report results without graphical interpretation.

1. Read g and compute its Euclidean magnitude.
2. Read C_K and compute $\kappa_F = 2 C_K / |g|$.
3. Construct a, b, and c from the three lengths and three angles using the canonical embedding equations.
4. Form all eight cell nodes by vector addition and compute the Gram matrix, face areas, normals, and volume.
5. Compute each rotor inertia from $I = \beta m r^2$.
6. Convert rpm to omega using $\omega = \text{rpm} \times 2 \pi \text{ divided by } 60$.
7. For the equal-energy state, solve outer omega from $E_o = E_i$ and evaluate both K-Field forces.
8. For the reaction state, set $L = I_i \omega_i$, solve outer $\omega = L/I_o$, and evaluate Equation (19).
9. Generate radius-ratio and mass-radius sweeps directly from Equations (11) and (23).
10. Verify energy equality in the prescribed state and angular-momentum equality in the reaction state.

Appendix E. Archival Record

Field	Record
Document title	K-Field Concentric Counter-Rotating Rotor Lateral-Force Analysis
Principal Investigator	P. Dwayne Esterline
Academic credentials	BS 1999 Huntington University; MBA 2008 Indiana Wesleyan University
Organization	Krytronx
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation UTC	2026-07-27T17:04:18Z
Generation epoch	1785171858
Output PDF	K-Field_concentric_rotor_lateral_force_analysis_1785171858.pdf
Machine-readable companion	K-Field_concentric_rotor_lateral_force_analysis_1785171858.json
Report generation	Python ReportLab; Helvetica, Helvetica-Bold, and Courier core fonts only
Figure generation	Python numerical computation from the exact ledger; raster figures embedded without font substitution
Coordinate frame for g	ICRS/J2000 Cartesian
Mechanical units	SI
Base-cell units	Native source length, area, and volume units

P. Dwayne Esterline
Principal Investigator

Krytronx

Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.